

MAMIDALA SAI ANUDEEP KUMAR

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PROFESSIONAL SUMMARY

B.Tech graduate with hands-on experience building Java/Spring Boot backend systems, including a distributed online judge with Redis and Docker-based microservices, and a secure REST API platform with SSRF protection. Strong in OOP, Problem Solving, and cloud fundamentals with AWS.

TECHNICAL SKILLS

Languages: Java, JavaScript, C++, SQL

Backend: Spring Boot, Spring MVC, Spring Security, REST APIs, Microservices Architecture, JWT, Redis

Databases: MySQL, PostgreSQL — schema design, indexing, query optimization

Tools: Docker, Docker Compose, Git/GitHub, Linux, Postman

Cloud: AWS (EC2, S3, Lambda) — hands-on practice deploying and configuring cloud resources

Frontend: React, HTML, CSS, Monaco Editor, Axios

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Problem Solving, SDLC, Agile

KEY PROJECTS

CodeForge — Distributed Online Judge | Java, Spring Boot, Microservices, Redis, Docker, REST APIs, React, MySQL, JWT | [GitHub](#)

- **Eliminated synchronous execution bottlenecks** using a Redis-backed async job queue and dedicated worker microservice, enabling non-blocking evaluation across 3 languages (Python, Java, C++) at scale.
- **Isolated 100% of code executions** via Docker containers and Java ProcessBuilder, blocking host system access and cross-submission interference.
- **Built a 4-stage real-time status pipeline** (PENDING → RUNNING → PASSED/FAILED/ERROR) surfacing testcase-level results — output, pass/fail, execution time — on a React + Monaco Editor frontend.
- **Secured 2 user roles (Admin/User)** end-to-end with JWT authentication, role-based access control, and Docker-enforced process isolation.

ProbeAPI — API Testing & Management Platform | Java, Spring Boot, Spring Security, JWT, PostgreSQL/MySQL, REST APIs | [GitHub](#)

- **Shipped 18 REST endpoints** across a 4-layer architecture (Controller → Service → Repository → DB), covering full CRUD for templates, collections, and paginated history.
- **Blocked SSRF attacks across 4 private IP ranges** (127.x, 192.168.x, 10.x, 172.16–31.x) plus IPv6 link-local, via a custom DNS-resolving SecurityValidator.
- **Isolated data across 5 relational tables** by scoping every DB query to the authenticated user through JWT claims — zero-trust by design.
- **Hardened validation with 4 layers of checks** — URL format, JSON syntax, 10MB payload limits, and HTTP method allowlisting — cutting invalid request surface area.

EDUCATION

B.Tech in Information Technology — St. Martin's Engineering College 2022 – 2026
CGPA: 7.3 / 10

Intermediate (Class XII) — Narayana Junior College 2020 – 2022
Percentage: 85.1%

ACHIEVEMENTS & CERTIFICATIONS

- **Round 2 qualifier, TCS CodeVita Season 13** — a global competitive programming contest.
- **170+ problems solved on LeetCode** across arrays, trees, graphs, and dynamic programming.
- **Finalist**, St. Martin's Engineering College Internal Hackathon.